

Cumulative Reflection

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On my first day at Iowa State, I remember the feeling of a fresh start, with both an overwhelming sense of excitement and nervousness. As a senior, I proudly walk the halls of Coover, ready to learn more about Hardware and Artificial intelligence while also bringing up underclassmen by teaching them computer architecture to the best of my ability. My time at Iowa State University has transformed me from a student who only knew basic programming into a well-rounded computer engineer who designs advanced hardware and software capable of sustainable technology while collaborating across disciplines. I strive to be a lifelong learner, tackling real challenges with ethical awareness and technical expertise. I've learned from my communications course proper workplace interdisciplinary and ethical communication practices. I've also experienced ethics courses, pointing out the potential pitfalls that unethical practices lead to. Most importantly to myself is my ability to learn rapidly while also applying said skills as I learn.

From theoretical math courses to concrete systems like digital logic, I have learned a multifaceted amount of information. I remember the first course that made me excited to be a computer engineer was CPRE 2810, digital logic, where I got to do digital design and experienced having a great lab partner who was dependable. Even in my non-engineering courses like English 3140, I had extraordinarily strong partners, and I really hope that where I end up next will have the same culture Iowa State has around work. I have recently accepted an internship position for the summer at boisei labs, a 'tech for good' venture studio. I am jubilant that I will be able to contribute to a group that shares my environmental values and meets the structure I wish for. This opportunity is possible because of my diverse learning experience at Iowa State University.

I think the best exposure to solving engineering problems in a collaborative setting was the graduate-level course I took as a junior, CPRE 5870, Hardware Design for Machine Learning. I was assigned a strong partner as well, but the challenges we faced, making a hardware accelerator for an AI model, were daunting and often felt impossible. The pride I have in my team cannot be understated, as not only were we one of the few, if not the only, teams to both finish all the labs as well as successfully complete and demonstrate our final project early. The design process was rigorous, there were no shortcuts, and yet our shared grit and determination allowed us to formulate and solve this problem.

Beyond the classroom, I implemented several of the skills I learned in my classes for personal projects. I have created my portfolio website from scratch using React and JavaScript, which I learned in my web design course. Even beyond that, I was able to participate as an undergraduate researcher at ARA, working on open-source 5G networking. I was tasked with making an existing open-source 5G software work with our outdoor hardware, which led me to

work with graduate students closely to enable me to complete the project. I have learned a lot from these graduate students, especially how to tackle complicated real-world problems.

As a member of Chip Forge, I have had the pleasure of meeting some of the smartest people I know. The tools are industry standard, which took a lot of effort to learn, but were well worth it. I am hoping that, as part of my senior design meeting, I can even get our chip fabricated and used in real-world settings. This learning was adapted and used instantaneously on the spot, leading to tangible results beyond just a grade in a classroom.

While I am a very capable engineer, I have had some major hiccups that required rapid adaptation. One of these problems was in one of my electrical engineering courses, in which I had an absent lab partner. This caused a lot of frustration as I had to notice his slack, and after 2 labs and me messaging him often, he finally informed me that he had dropped the course. While this could be devastating, I was able to link up with another team and successfully collaborate, and direct us to successfully complete our labs.

Knowing what I know now as a senior, I would have taken my time to complete my program. I am graduating a semester early, which is fantastic, but I may have missed some entrepreneurial club activities I might have otherwise participated in. I have attempted to participate, but my high credit load and choice of difficult courses forced me to stay focused on my courses. My experience gained is unparalleled, but taking the time to explore other passions could have improved my general abilities.

Part of my studies has been focused on artificial intelligence. The main course I took for this, mentioned earlier, CPRE 5870, Hardware Design for Machine Learning, forced me to undergo a very strenuous learning cycle, particularly regarding model quantization. I had started with no machine learning knowledge and had to start at the basics of matrix multiplication and back propagation before learning some of the higher-level concepts. This initial gathering was particularly important for later steps, although the assimilation of this knowledge was exceedingly difficult and required multiple rereads, watching alternative sources, and even going to some office hours. After incorporating this knowledge, I was able to finally implement the proper quantization algorithms, succeeding after what was a week of being in the lab every single night. At the moment, it was incredibly frustrating to be struggling so much on an algorithm, even though everyone else in the course was also struggling with the same concept, but the payoff was unparalleled, and now I know advanced machine learning strategies and optimizations.

My time at Iowa State University forced me to learn new strategies for learning content. I have discovered that just going to class was not going to cut it. The portability of electronic notes was a lifesaver, along with online videos from content creators like The Organic Chemistry Tutor, who covered math and science subjects in a concise and understandable way. The advent of Generative Large Language Models like ChatGPT has been disruptive to say the least. I personally avoided using these models, especially in an academic setting the first two and a half

years. I had changed my mind when I had it explain certain algorithms for solving complex circuit problems. Despite its utility in tutoring, I actively made an effort to avoid using it beyond receiving feedback for my work and as a virtual tutor. While this works great for me, helping me learn significantly faster, I do worry for many of my colleagues who, in both collaborative settings and as students during my tenure as an undergraduate TA, have shown a lack of understanding of core concepts and a heavy reliance on these AI models.

I look brightly towards my future; I anticipate that my touch at boisei labs will be impactful for everyday people while pushing us towards a cleaner planet. I plan to deepen my understanding of AI, keeping up to date with research papers regarding the subject and attempting to contribute to creating an AI-powered future that is sustainable. The sustainability of artificial intelligence is questionable, but it does not need to be given proper consideration. I believe the best outcome with artificial intelligence is a post-scarcity society, at least scarcity for essentials. Even if AI cannot scale, in its current state, it has a niche in the market and can be a powerful productivity booster.

Iowa State has profoundly changed me in the best way possible, from an experienced yet enthusiastic young adult into a well-rounded engineer, enthusiastic about the hardware and software problems to be solved. I have experienced a wide variety of different courses from math and science to the history of American Rock and Roll. Iowa State has given me the tools, work ethic, and abilities to go beyond being just a competent computer engineer into being a responsible innovator who will shape technology for the betterment of society and the planet.